

Cauchy's Integral Theorem

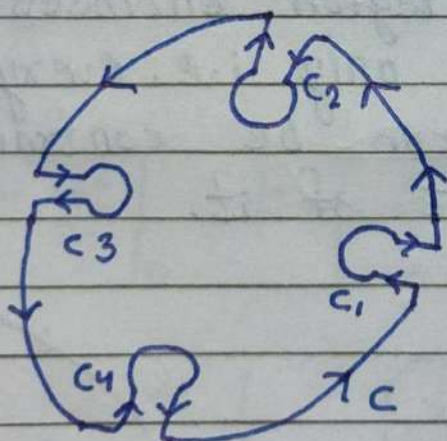
If $f(z)$ is analytic and $f'(z)$ is continuous within and on a simple closed curve C then,

$$\oint_C f(z) dz = 0.$$

Extension of Cauchy's Integral Theorem for Multiply Connected Region,

If $f(z)$ is analytic between simple closed curve C and $C_1, C_2, C_3, \dots, C_n$, then

$$\oint_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz + \dots + \int_{C_n} f(z) dz$$

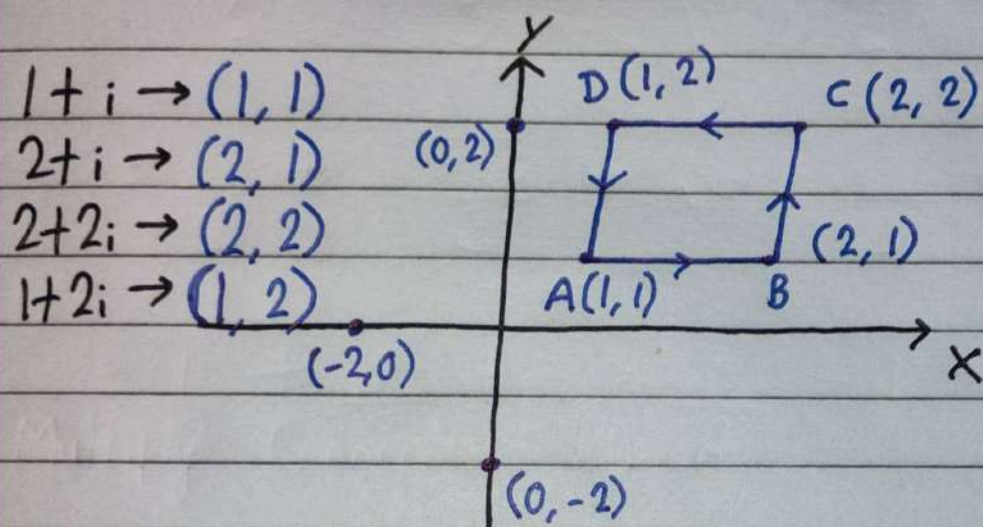


Q.1) Evaluate $\oint_C \frac{2z^2+5}{(z+2)^3(z^2+4)} dz$, where C is a

square with vertices at $1+i$, $2+i$, $2+2i$ & $1+2i$.

Solⁿ:

$$f(z) = \frac{2z^2+5}{(z+2)^3(z^2+4)}$$



For singular points (Points at which $f(z)$ is not analytic),

$$(z+2)^3(z^2+4) = 0$$

$$\Rightarrow (z+2)^3 = 0 \quad \text{OR} \quad \Rightarrow (z^2+4) = 0$$

$$\Rightarrow z = -2$$

$$\Rightarrow z^2 = -4$$

$$\Rightarrow z = \pm\sqrt{-4}$$

$$\Rightarrow z = \pm 2i$$

$$z = -2 \rightarrow (-2, 0)$$

$$z = 2i \rightarrow (0, 2)$$

$$z = -2i \rightarrow (0, -2)$$

$\therefore f(z)$ is analytic inside the square.

Therefore, by Cauchy's Integral Theorem,

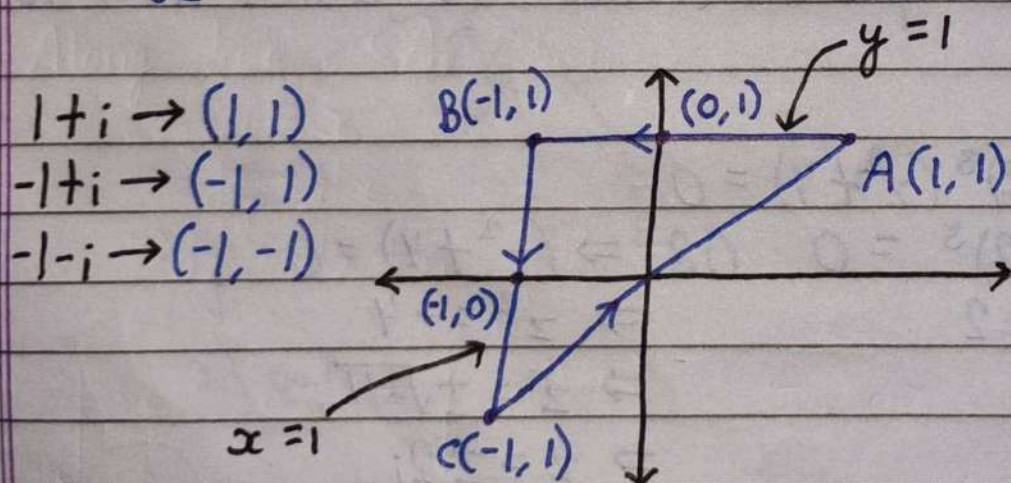
$$\oint_C \frac{2z^2 + 5}{(z+2)^3(z^2+4)} dz = 0$$

Q.2) Verify Cauchy's Integral Theorem by integrating e^{iz} along the boundary of the triangle within the vertices $1+i$, $-1+i$ & $-1-i$.

Solⁿ:

Here, $f(z) = e^{iz}$, which is always analytic.

So, $\oint_C f(z) dz = 0$ where C is a triangle.



Along line AB ,

$$\begin{aligned} y &= 1 \\ \Rightarrow dy &= 0 \\ \Rightarrow dz &= dx + i dy = dx \end{aligned}$$

$$\int_{AB} e^{iz} dz = \int e^{i(x+iy)} (dx + i dy)$$

$$= \int_{-1}^1 e^{i(x+i)} dx$$

$$= \int_{-1}^1 e^{ix} \cdot e^{-1} dx$$

$$= e^{-1} \int_{-1}^1 e^{ix} dx$$

$$= e^{-1} \left[\frac{e^{ix}}{i} \right]_{-1}^1$$

$$= \frac{e^{-1}}{i} [e^i - e^{-i}]$$

$$= -i \cdot e^{-1} (e^i - e^{-i})$$

$$\left[\frac{1}{i} = -i \right]$$

✓ $\int_{AB} e^{iz} dz = -i e^{-1-i} + i e^{i-1}$

Along line BC,

$$x = -1$$

$$\Rightarrow dx = 0$$

$$\Rightarrow dz = dx + i dy = i dy$$

$$\int_{BC} e^{iz} dz = \int_1^{-1} e^{i(-1+iy)} i dy$$

$$= i \int_1^{-1} e^{-i-y} dy$$

$$= i e^{-i} \int_1^{-1} e^{-y} dy$$

$$= i e^{-i} [-e^{-y}]_1^{-1}$$

$$= i e^{-i} [-e^1 + e^{-1}]$$

$$\checkmark \int_{BC} e^{iz} dz = -i e^{-i+1} + i e^{-i-1}$$

Along line CA,

$$\begin{aligned} y &= x \\ \Rightarrow dy &= dx \\ \Rightarrow dz &= dx + i dy = (1+i) dx \end{aligned}$$

$$\int_{CA} e^{iz} dz = \int_{-1}^1 e^{i(1+i)x} (1+i) dx$$

$$= (1+i) \int_{-1}^1 e^{i(1+i)x} dx$$

$$= \cancel{(1+i)} \left[\frac{e^{i(1+i)x}}{\cancel{(1+i)}i} \right]_{-1}^1$$

$$= \left[\frac{e^{i(1+i)x}}{i} \right]_{-1}^1$$

$$= \frac{e^{i(1+i)} - e^{i(-1-i)}}{i}$$

$$= -i (e^{i-1} - e^{-i+1})$$

$$\int_{CA} e^{iz} dz = -ie^{i-1} + ie^{-i+1}$$

Therefore,

$$\oint_C e^{iz} dz = \int_{AB} e^{iz} dz + \int_{BC} e^{iz} dz + \int_{CA} e^{iz} dz$$

$$\Rightarrow \oint_C e^{iz} dz = \cancel{-ie^{i-1} + ie^{-i+1}} - \cancel{ie^{i+1} - ie^{-i-1}} + \cancel{ie^{i-1} - ie^{-i+1}}$$

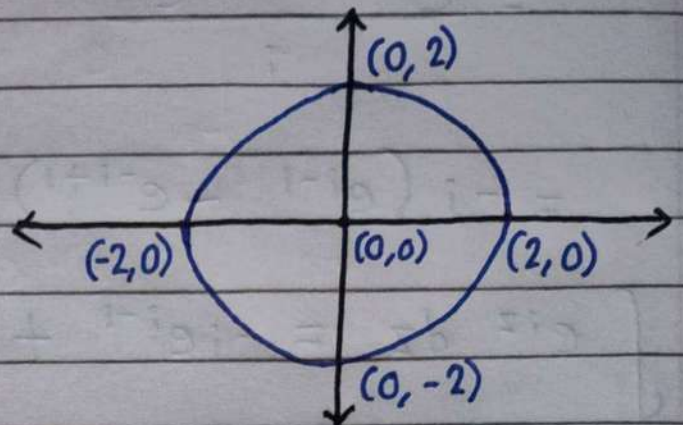
$$\Rightarrow \oint_C e^{iz} dz = 0.$$

Hence, the Cauchy's Integral Theorem is verified.

Q.3) Evaluate $\oint_C \frac{e^z}{z^2+9} dz$, where C is $|z|=2$.

Solⁿ:

$$f(z) = \frac{e^z}{z^2+9}$$



Points of singularity,

$$z^2+9=0$$

$$\Rightarrow z^2 = -9$$

$$\Rightarrow z = \sqrt{-9}$$

$$\Rightarrow z = \pm 3i$$

$$z = 3i \rightarrow (0, 3)$$

$$z = -3i \rightarrow (0, -3)$$

These points lie outside C .

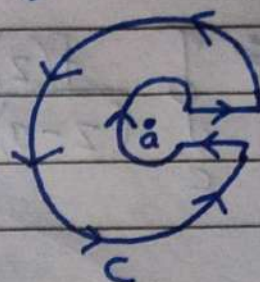
$\therefore f(z)$ is analytic inside C .

$$\therefore \oint_C \frac{e^z}{z^2+9} dz = 0. \text{ [By Cauchy's Integral]}$$

Cauchy's Integral Formula

If $f(z)$ is analytic within and on a simple closed curve C and 'a' is any point within C then,

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-a)} dz$$



Theorem:

If a function $f(z)$ is analytic in a domain D , then at any point $z=a$ of D , $f(z)$ has derivatives of all orders all of which are again analytic functions in D , their value is given by,

$$f^n(a) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z-a)^{n+1}} dz$$

when $n=0$,

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-a)} dz$$

$$\Rightarrow \oint_C \frac{f(z)}{(z-a)} dz = 2\pi i f(a)$$

when $n=1$,

$$f'(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-a)^2} dz$$

$$\Rightarrow \oint_C \frac{f(z)}{(z-a)^2} dz = 2\pi i f'(a).$$

Q.1) Evaluate $\oint_C \frac{e^{-z}}{(z+1)} dz$, where C is a circle $|z|=2$.

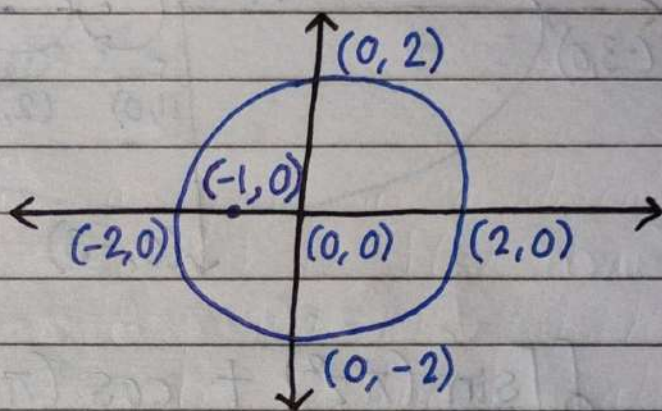
Solⁿ:

Given, $\oint_C \frac{e^{-z}}{(z+1)} dz$

Comparing above with $\oint_C \frac{f(z)}{(z-a)} dz$, we get,

$$f(z) = e^{-z}$$

$$a = -1$$



$\therefore f(z) = e^{-z}$ is always analytic and $a = -1$ is lying inside C .

$\therefore$ By Cauchy's Integral Formula,

$$\oint_C \frac{e^{-z}}{(z+1)} dz = 2\pi i \cdot f(-1)$$

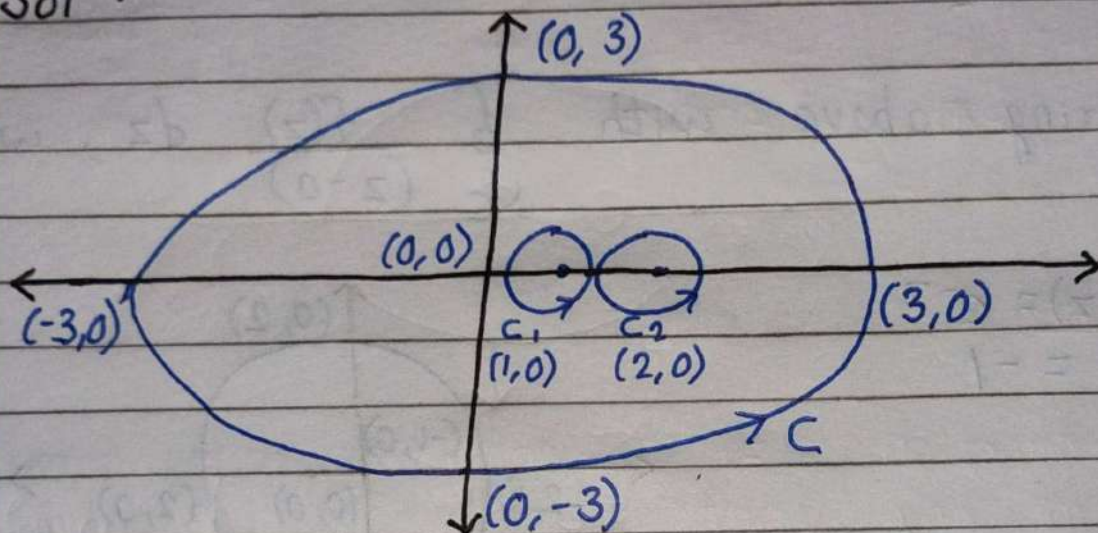
$$= 2\pi i (e^{-z})$$

$$= 2\pi i e \quad [z = -1]$$

Q.2) Evaluate $\oint_C \left[\frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)(z-2)} \right] dz$,

where C is a circle $|z| = 3$.

Solⁿ:



$$\oint_C \left[\frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)(z-2)} \right] dz$$

$$= \int_{C_1} \left[\frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-2)} \right] dz + \int_{C_2} \left[\frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)} \right] dz$$

Now,

$$\int_{C_1} \left[\frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-2)} \right] dz = 2\pi i f(1)$$

Here, $f(z) = \frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-2)}$

& $a = 1$

$$f(1) = \frac{\sin \overset{0}{\pi} + \cos \overset{-1}{\pi}}{(-1)} = \frac{-1}{-1} = 1$$

$$\therefore \int_{C_1} \left[\frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-2)(z-1)} \right] dz = 2\pi i$$

Also,

$$\int_{C_2} \left[\frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)(z-2)} \right] dz = 2\pi i f(2)$$

Here, $f(z) = \frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)}$

& $a = 2$

$$f(2) = \frac{\sin \overset{0}{4\pi} + \cos \overset{1}{4\pi}}{1} = 1$$

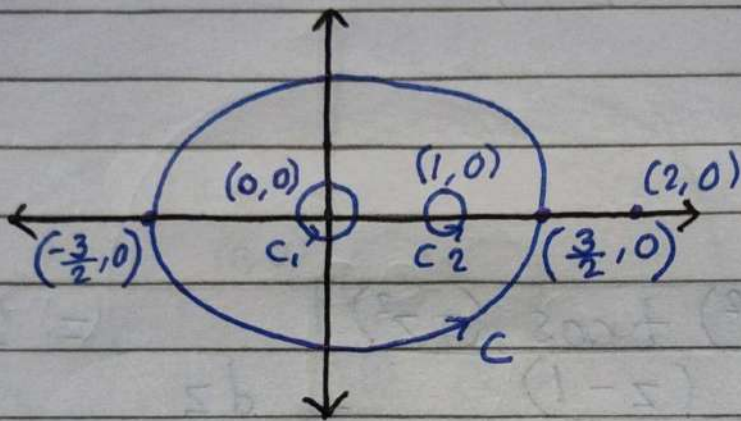
$$\therefore \int_{C_2} \left[\frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)(z-2)} \right] dz = 2\pi i$$

Hence, $\oint_C \left[\frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)(z-2)} \right] dz = 4\pi i$

Q.3) Evaluate $\oint_C \frac{4-3z}{z(z-1)(z-2)} dz$, where C is a

circle $|z| = 3/2$.

Solⁿ:



Given, $\oint_C \frac{4-3z}{(z-0)(z-1)(z-2)} dz$

Here, $z(z-1)(z-2) = 0$
 $\Rightarrow z = 0, 1, 2$

$\therefore z = 0, 1$ lies inside C .

$\therefore \oint_C \frac{4-3z}{z(z-1)(z-2)} dz$

$$= \int_{C_1} \left[\frac{4-3z}{(z-1)(z-2)} \right] dz + \int_{C_2} \left[\frac{4-3z}{z(z-2)} \right] dz$$

Now,

$$\int_{C_1} \left[\frac{4-3z}{(z-1)(z-2)} \right] \frac{dz}{z} = 2\pi i f(0)$$

$$f(0) = \frac{4-3 \times 0}{(0-1)(0-2)} = \frac{4}{(-1)(-2)} = 2$$

$$\text{So, } \int \left[\frac{4-3z}{(z-1)(z-2)} \right] \frac{dz}{z} = 2\pi i \times 2 = 4\pi i$$

Also,

$$\int_{C_2} \left[\frac{4-3z}{z(z-2)} \right] \frac{dz}{(z-1)} = 2\pi i f(1)$$

$$f(1) = \frac{4-3 \times 1}{1(1-2)} = \frac{1}{(-1)} = -1$$

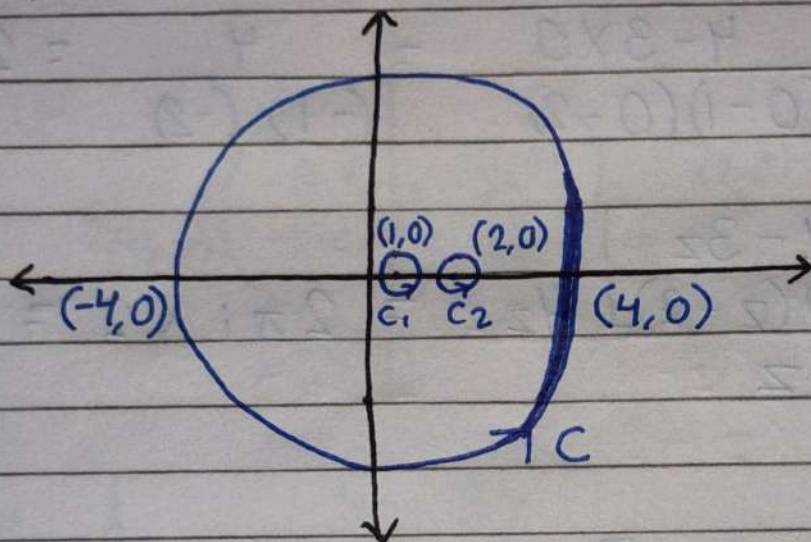
$$\text{So, } \int \left[\frac{4-3z}{z(z-2)} \right] \frac{dz}{(z-1)} = 2\pi i \times (-1) = -2\pi i$$

$$\therefore \oint_C \frac{4-3z}{z(z-1)(z-2)} dz = 2\pi i.$$

Q.4) Evaluate $\oint_C \frac{\sin(\pi z) + \cos(\pi z)}{(z-1)(z-2)} dz$, where

C is a circle $|z| = 4$.

Solⁿ:



$$\oint_C \frac{\sin(\pi z) + \cos(\pi z)}{(z-1)(z-2)} dz$$

$$= \int_{C_1} \left[\frac{\sin(\pi z) + \cos(\pi z)}{(z-2)} \right] dz + \int_{C_2} \left[\frac{\sin(\pi z) + \cos(\pi z)}{(z-1)} \right] dz$$

Now,

$$\int_{C_1} \left[\frac{\sin(\pi z) + \cos(\pi z)}{(z-2)} \right] dz = 2\pi i f(1)$$

$$f(1) = \frac{\sin \overset{0}{\pi} + \cos \overset{-1}{\pi}}{1-2} = \frac{-1}{-1} = 1$$

$$\text{So, } \int_{C_1} \left[\frac{\sin(\pi z) + \cos(\pi z)}{(z-2)(z-1)} \right] dz = 2\pi i$$

Also,

$$\int_{C_2} \left[\frac{\sin(\pi z) + \cos(\pi z)}{(z-1)(z-2)} \right] dz = 2\pi i f(2)$$

$$f(2) = \frac{\sin \overset{0}{(2\pi)} + \cos \overset{1}{(2\pi)}}{2-1} = \frac{1}{1} = 1$$

$$\text{So, } \int_{C_2} \left[\frac{\sin(\pi z) + \cos(\pi z)}{(z-1)(z-2)} \right] dz = 2\pi i$$

$$\therefore \oint_C \frac{\sin(\pi z) + \cos(\pi z)}{(z-1)(z-2)} dz = 4\pi i.$$

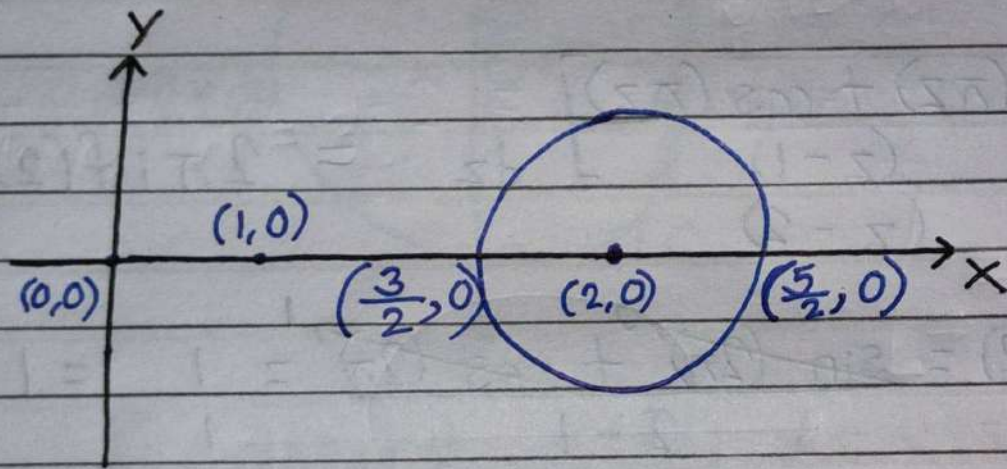
Q.5) Evaluate $\oint_C \frac{z}{z^2 - 3z + 2} dz$, where C is a

circle $|z - 2| = 1/2$.

$$|z - (2 + i \cdot 0)|$$

Solⁿ:

Here, centre of circle is $(2, 0)$
& radius is $1/2$.



For singularity,

$$\begin{aligned} z^2 - 3z + 2 &= 0 \\ \Rightarrow z^2 - (2+1)z + 2 &= 0 \\ \Rightarrow z^2 - 2z - z + 2 &= 0 \\ \Rightarrow z(z-2) - 1(z-2) &= 0 \\ \Rightarrow (z-1)(z-2) &= 0 \\ \Rightarrow z = 1, 2 \end{aligned}$$

$\therefore z = 2$ lies inside C .

$$\therefore a = 2$$

$$\text{So, } \oint_C \frac{z}{z^2 - 3z + 2} dz$$

$$= \oint_C \frac{z}{(z-1)(z-2)} dz$$

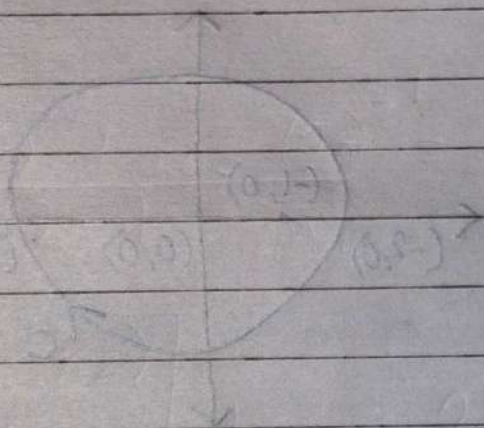
$$= \oint_C \frac{\left[\frac{z}{z-1} \right]}{(z-2)} dz$$

$$= 2\pi i \cdot f(2)$$

$$= 2\pi i \cdot \left(\frac{2}{2-1} \right)$$

$$= 2\pi i \cdot 2$$

$$= 4\pi i$$



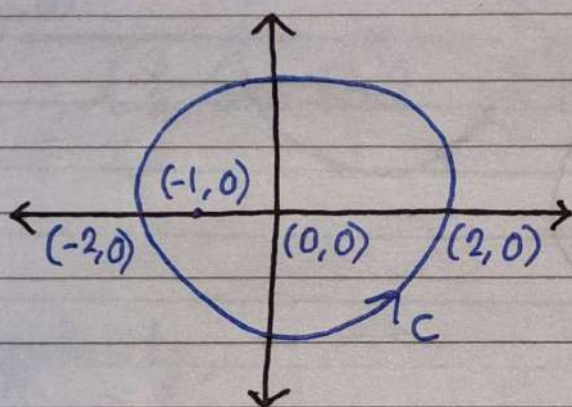
Q.6) Use Cauchy Integral formula to evaluate,

$$\oint_C \frac{e^{2z}}{(z+1)^4} dz \text{ where } C \text{ is the circle } |z|=2.$$

Solⁿ:

For given, using formula $\oint_C \frac{f(z)}{(z-a)^{n+1}} dz = \frac{2\pi i}{n!} f^n(a)$

Here, $n=3$.



$$\left[\begin{array}{l} z+1=0 \\ \Rightarrow z=-1 \Rightarrow (-1,0) \end{array} \right]$$

$\therefore z=-1$ lies inside C and e^{2z} is analytic inside C .

$$\oint_C \frac{e^{2z}}{(z+1)^4} dz = \frac{2\pi i}{3!} f'''(-1)$$

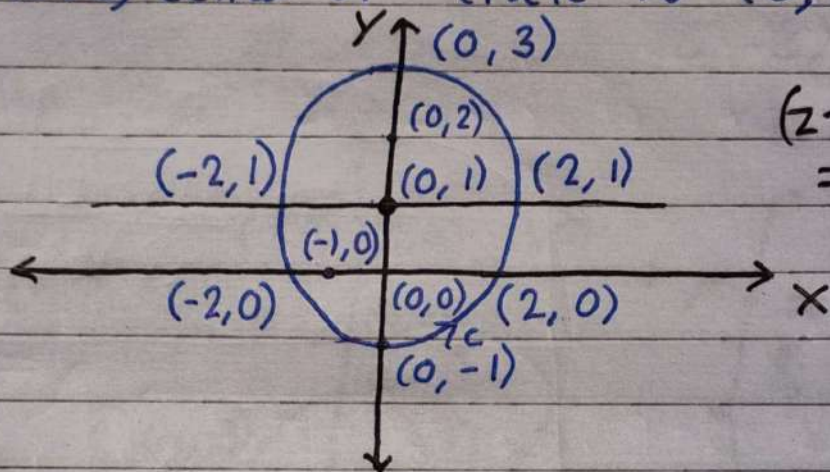
$$\begin{aligned} f(z) &= e^{2z} \\ \Rightarrow f'''(z) &= 8 \cdot e^{2z} \\ \Rightarrow f'''(-1) &= 8 \cdot e^{-2} = \frac{8}{e^2} \end{aligned}$$

$$\Rightarrow \oint_C \frac{e^{2z}}{(z+1)^4} dz = \frac{2\pi i}{6} \cdot \frac{8}{e^2} = \frac{8\pi i}{3e^2}$$

Q.7) Use Cauchy Integral Formula to evaluate,

$$\oint_C \frac{(z-1)}{(z+1)^2(z-2)} dz, \text{ where } C \text{ is the circle } |z-i|=2.$$

Solⁿ: Here, centre of circle is $(0,1)$ & radius = 2.



$$(z+1)^2(z-2)=0 \\ \Rightarrow z = -1, 2$$

$\therefore z = -1$ lies inside C .

$\therefore a = -1$.

$f(z) = \frac{(z-1)}{(z-2)}$ is analytic inside C .

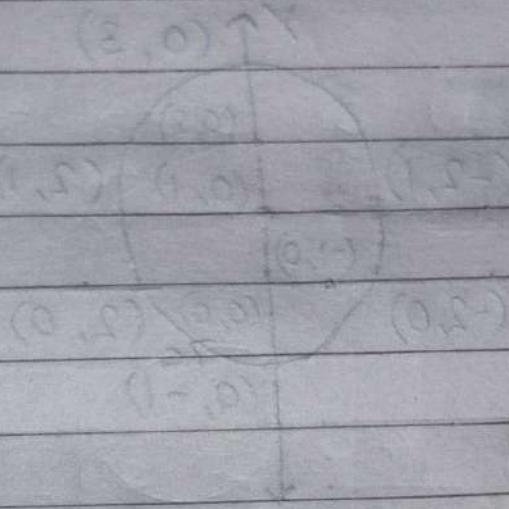
$$\text{So, } \oint_C \frac{(z-1)}{(z+1)^2(z-2)} dz$$

$$= \oint_C \left[\frac{(z-1)}{(z-2)} \right] \frac{1}{(z+1)^2} dz = \frac{2\pi i}{1!} f'(-1)$$

$$f(z) = \frac{z-1}{z-2} \Rightarrow f'(z) = \frac{-1}{(z-2)^2}$$

$$f'(-1) = -\frac{1}{9}$$

$$\therefore \oint_C \frac{(z-1)}{(z+1)^2(z-2)} dz = -\frac{2\pi i}{9}$$

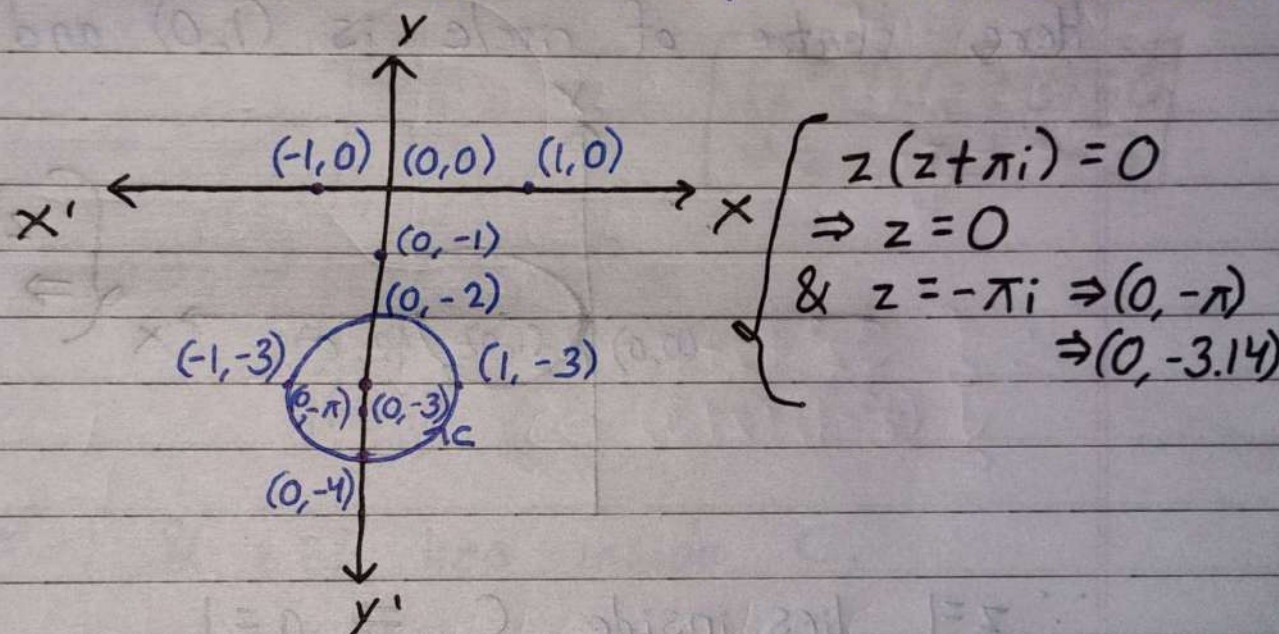


Q.8) Evaluate by Cauchy Integral formula,

$$\oint_C \frac{dz}{z(z+\pi i)} \text{ where } C \text{ is a circle } |z+3i|=1.$$

Solⁿ:

Given, centre of circle is $(0, -3)$ & radius is 1.



$\therefore z = -\pi i$ lies inside C .

$$\therefore a = -\pi i$$

$$\Rightarrow z - a = z + \pi i$$

$$\oint_C \frac{\left(\frac{1}{z}\right)}{(z + \pi i)} dz = 2\pi i f(-\pi i) = 2\pi i \left(\frac{-1}{\pi i}\right) = -2.$$

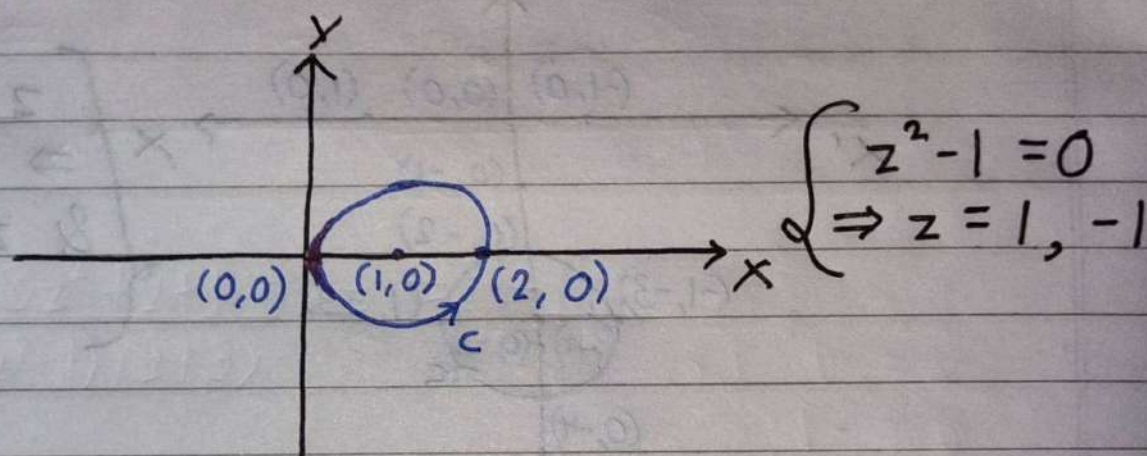
$\left[f(z) = \frac{1}{z} \text{ is analytic inside } C \right]$

Q.9) Evaluate by Cauchy Integral formula,

$$\oint_C \frac{3z^2+z}{(z^2-1)} dz, \text{ where } C \text{ is the circle } |z-1|=1.$$

Solⁿ:

Here, centre of circle is $(1, 0)$ and radius is 1.



$\therefore z=1$ lies inside $C. \Rightarrow a=1$

$$\text{So, } \oint_C \frac{z(3z+1)}{(z-1)(z+1)} dz \quad \left[f(z) \text{ is analytic inside } C. \right]$$

$$= \int \left[\frac{3z^2+z}{(z+1)(z-1)} \right] dz = 2\pi i \cdot f(1)$$

$$f(1) = \frac{3+1}{1+1} = \frac{4}{2} = 2$$

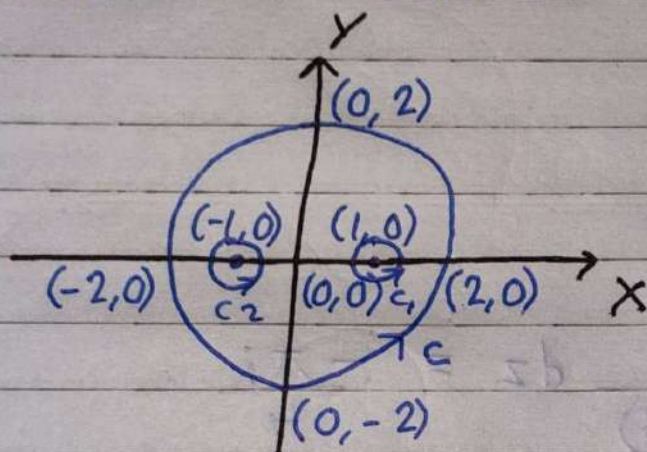
$$\therefore \oint_C \frac{3z^2+z}{(z^2-1)} = 4\pi i.$$

Q.10) Evaluate by Cauchy Integral formula,

$$\oint_C \frac{3z^2 + z + 1}{(z^2 - 1)(z + 3)} dz, \text{ where } C \text{ is the circle } |z| = 2.$$

Solⁿ:

Here, centre of circle is $(0, 0)$ & radius 2.



$$\begin{aligned} (z^2 - 1)(z + 3) &= 0 \\ \Rightarrow z &= -1, 1, -3 \end{aligned}$$

$\therefore z = -1$ & $z = 1$ lies inside C .

$$\text{So, } \oint_C \frac{3z^2 + z + 1}{(z - 1)(z + 1)(z + 3)} dz$$

$$= \oint_{C_1} \left[\frac{3z^2 + z + 1}{(z + 1)(z + 3)} \right] dz + \oint_{C_2} \left[\frac{3z^2 + z + 1}{(z - 1)(z + 3)} \right] dz$$

$$= 2\pi i \cdot f(1) + 2\pi i \cdot f(-1)$$

$$= 2\pi i \left(\frac{3 + 1 + 1}{(1 + 1)(1 + 3)} \right) + 2\pi i \left(\frac{3 - 1 + 1}{(-1 - 1)(-1 + 3)} \right)$$

$$= 2\pi i \cdot \frac{5}{8} + 2\pi i \cdot \frac{3}{-4}$$

$$= \frac{5\pi i}{4} - \frac{3\pi i}{2}$$

$$= \frac{5\pi i - 6\pi i}{4}$$

$$= -\frac{\pi i}{4}$$

$$\therefore \oint_C \frac{3z^2 + z + 1}{(z^2 - 1)(z + 3)} dz = -\frac{\pi i}{4}$$

